



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

Submitted by:

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CEP Mapping:

1. Knowledge Profile (K's) Mapping

K's	Attribute	How K's Are Addressed Through Our Project	CO	PO
K4	Specialist Knowledge	Used Arduino UNO, ESP32-CAM, motor driver, sensors (DHT11/DHT22, MQ, Dust, Ultrasonic, IR) with WiFi.	CO1, CO2	PO1
K5	Engineering Design	Designed the system with ESP32-CAM, Arduino UNO, motors, motor driver, and environmental & obstacle detection sensors.	CO3, CO4	PO3, PO5
K6	Engineering Practice	Built it using Arduino IDE, ESP32 libraries, sensors, obstacle avoidance algorithms and WiFi.	CO4, CO5	PO4, PO5, PO6
K7	Comprehension	Made a simple IoT dashboard/app to control the car, monitor live	CO5	PO7

		video and environmental + obstacle data.		
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2. Complex Engineering Problems (P's) Mapping

P's	Attribute	How P's Are Addressed Through Our Project	CO	PO
P1	Depth of Knowledge Required	Learned integration of ESP32-CAM, Arduino, WiFi, motor driver, and sensors, obstacle detection modules.	CO1, CO2, CO4, CO5	PO1, PO2, PO7
P3	Depth of Analysis Required	Tested WiFi range, video latency, sensor range & accuracy, distance thresholds, motor response, environmental accuracy and power management.	CO1, CO2, CO3	PO1, PO2, PO6
P7	Interdependence	Integrated ESP32-CAM, Arduino, motor driver,	CO1, CO3, CO5	PO1, PO3, PO7

		motors, sensors, and battery for complete system.		
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3. Complex Engineering Activities (A's) Mapping

A's	Attribute	How A's Are Addressed Through Our Project	CO	PO
A1	Range of Resources	Used Arduino UNO, ESP32-CAM, motor driver, ultrasonic/IR sensors, environmental sensors, chassis, battery, IDE, libraries and WiFi.	CO1, CO4	PO1, PO6, PO7
A4	Consequences for Society & Environment	Provides low-cost surveillance, monitors environmental safety, obstacle avoidance and uses rechargeable power batteries.	CO5, CO7	PO6, PO7